PRINCIPLES OF FISH BIOLOGY (ZOO603T)
CURRENT MIDTERM PAPER SPRING 2024(2JUNE)

SHORT QUESTIONS(3)

1. **Gill Rakers:** help in fish feeding. Gill rakers filter out food and debris, preventing them from entering the gills.

2. What is the Scientific name of Shovelnose sturgeons?

Shovelnose sturgeons (*Scaphirhynchus platorynchus*)

3. Jugular Fin

When pelvic fins are below the pectoral fins, they termed thoracic. When a thoracic pelvic fin is attached under gills, it may called jugular.

4. Physoclistic situation in fishes

In advanced teleosts, especially the acanthopterygians, the bladder has lost its connection with the digestive tract, a condition called physoclistic.

5. Three parts of fish brain

Fish brains are generally divided into general regions. •

1) Prosencephalon,
2) Mesencephalon and
3) rhombencephalon are the regions of brain.

6. Types Of Scales

There are four main kinds of scales and numerous variations of each kind.

(i) Placoid (sharks and rays)
(ii) Cosmoid (lungfishes and some fossil fishes)
(iii) Ganoid (bichirs , Bowfin, paddlefishes, gars, sturgeons)
(iv) Cycloid and Ctenoid (most bony fishes)

7. Systematics and Taxonomy:

Systematics is the study of the diversity of fish and the evolutionary relationships between these organisms

Taxonomy is the scientific study of naming, defining and classifying groups of biological organisms based on shared characteristics

8. Importance of Taxonomy and systematics

□ Sound systematic data are important for conservation. Without accurate taxonomy, It is impossible to identify the species and evaluate their conservation status, □ It is impossible to properly manage their fisheries, □ it is impossible to evaluate the conservation value of habitats or areas, ➢ it is impossible to establish strategies ➢ it is impossible to set priorities.

9. Taxonomic characters

• Taxonomic Characters, used to classify organisms on the basis of Similarities and differences.

1) ***Morphometric characters:*** • used for measureable structures
2) ***Meristic characters:*** • used for any countable structures
3) ***Molecular characters:***• nuclear DNA and Mitochondrial DNA
4) ***Anatomical character:***• skeleton and characters of the soft anatomy

10. Names of Herbivorous fishes.

Species of these herbivorous fish include: ***parrotfish***, which eat corals that break Off the reef and excrete them as white sand; ***damselfish***, which fend off the Macro algae growth that kills coral; and ***surgeonfish***, whose most famous representative is ***Dory*** from Finding Nemo.

11. Parasites

• It is perhaps the most unusual and highly evolved feeding habit among animals. • In this case, one fish gets shelter and food from the other fish. •

Examples: Some parasitic fish, such as lamprey (Petromyzonidae) and hagfish (Myxinidae), make holes in the sides of the host's body to absorb body fluids by sucking. • Sea lamprey (Petromyzon marinus) of the western North Atlantic and adjacent continents live on the body of the whales as parasite and feed on host's body fluids by sucking.

12. Protrusible mouth

• A protrusible mouth allows a fish to extend its reach when attempting to snatch prey or food particles. • ***For example,*** kissing gourami uses its protrusible mouth to defend territory against others of the same species.

13. Define PUFA

It stands for polyunsaturated fatty acids, a type of fat that is essential for human health. This category includes both omega-3 and omega-6 fatty acids, which are important for brain health, heart health, and overall inflammation regulation in the body. Good sources of PUFA include fatty fish, nuts, seeds, and vegetable oils.

14. What is caudal peduncle?

The caudal peduncle is the base of the caudal fin. Peduncle means stem, and the caudal peduncle is where the strong swimming muscles of the tail are found. Together, the caudal fin acts like a "propeller" for the fish, and the caudal peduncle acts like a motor.

15. Pufferfish poisoning

• Tetrodotoxic poisoning occurs following the ingestion of pufferfish, boxfish or porcupine fish. • This is also known as puffer or fugu poisoning. • The poisoning is due to toxin known as tetrodotoxin which is concentrated in the internal organs such as liver, ovary and intestines. • The skin also contains some glands which secrete these toxins. • Usually the body muscles are free of the toxin.

16. How barbles are classified?

• A barbel is a slender, whisker-like sensory organ near the mouth. • Fish that have barbels include the catfish, the carp, the goatfish, the hagfish, the sturgeon, the zebrafish, the black dragonfish and some species of shark such as the sawshark.

Barbels can be classified as:

• Tender barbels, rigid barbels, barbels without taste buds, with taste buds and without cartilage (Cyprinidae, Carps). • Barbels with taste buds and cartilage (Catfish, Goatfish) and with an axial rod of striated muscle (Polymixia type)

17. Define (cannibalistic, piscivores)

Cannibalistic: • Cannibalism is the act of consuming another individual Of the same species as food. ***Examples:*** angel fish, gold fish.

Piscivores: • A piscivore is a carnivorous animal that eats primarily fish. ***Examples:*** barracuda, lionfish.

18. Detritivores

• – Fish consume decaying plants and animals matter as food. • – 10 percent of all fish species • – feed on decomposing organic matter • Detritivores may also obtain nutrition by coprophagy, which is a feeding strategy involving the consumption of feces. • ***Examples*** • – lobsters • – fiddler crabs

19. Larvivores' fish

• – Fish exclusively consume insect larvae as food. • – Larvivores are specialized on only one type of food

• – and are referred to as; • – Larvivorus • – insectivores • – mollusks feeders • – piscivorous and • – Crustacean's eaters.

Larvivorous fish are those that feed on immature stages of mosquitoes. According to Job15, larvivores' fish must be small, hardy and capable of getting about easily in shallow waters among thick weeds where mosquitoes find suitable breeding grounds.• ***Recommended food*** for the Larvivores would be: • Mosquito larvae and pupae

• ***Examples*** • – dwarf gourami fish • – Dwarf snakehead fish • – spotted snakehead • – striped snakehead

20. Role of fish to reduce parasite?

. • Some fish species eat mosquito larvae and pupae. In disease control policy documents, the World Health Organization (WHO) includes biological control of malaria vectors by stocking ponds, rivers,

and water collections near where people live with larvivorous fish to reduce Plasmodium parasite transmission.

Example: Fishes like Gambusia are introduced into ponds to control the spread of malaria

21. How gills are protected in fishes?

Gills in fishes are protected by a combination of structures and mechanisms. The ***operculum***, a bony flap, covers and shields the gills from debris and predators. ***Gill rakers*** filter out food and debris, preventing them from entering the gills. A layer of ***mucus*** traps additional debris, while ***gill arches*** provide a supportive framework for the delicate gill filaments. The ***gill epithelium,*** a layer of cells, regulates gas exchange and maintains the integrity of the gill tissue. These protective measures enable fishes to efficiently extract oxygen from the water while keeping their respiratory surfaces safe from harm.

22. Ray finned fish/Why actinoptergyii are called ray finned fish?

Their fins lack muscular lobes that is why called as ray-finned fishes. • They usually possess swim bladders, gas filled sacs along dorsal wall of the cavity that regulate buoyancy. • Paired fins supported by dermal rays • Base of fins is not muscular • Tail fins almost equal to upper and lower lobes • Blind olfactory

Ray-finned fishes include herbivores, carnivores, omnivores, zooplankton (for example, small crustaceans such as copepods and fish larvae) consumers, and detritus feeders, or those that feed on dead organic matter.

23. Skeleton of a fish

• The skeletal system forms the framework of the body and is mainly composed of bones and tissues. The main functions of the skeletal system are as follows: • Protection. • Helps in movement. • Provides shape and support. • They are the site for the storage of minerals like calcium. • General features of fishes include fins, streamlines and scales and tails. But differences are highlighted under their skins. • Hence classification is much easier based on the skeletal system. • A variety of fishes is found in aquatic habitats Some of these fishes are cartilaginous fishes (Chondrichthyes) whereas others are bony fishes (Osteichthyes). • The skeletal system of fishes is either composed of thin and flexible cartilage or hard calcified bones or both. • They are good swimmers and their body structures are designed accordingly.

24. Characteristics of Cephalospidomorph

• Sucking mouth

• Teeth are present

• Rasping tongue

• Seven pairs of slits

• Blind olfactory sacs

• Cephalospidomorph were, like most contemporary fishes, very well armored.

• They also lacked a swim bladder, and would not have been able to keep afloat without actively Swimming.

• The head shield provided some lift though, and would have made the cephalaspidomorphi better swimmers than most of their contemporaries.

• The head shield was particularly well developed, protecting the head, gills and the anterior section of the viscera.

25. What happened if there is more plankton in ocean?

If there is an increase in the plankton population in the ocean, it can have various effects on the marine ecosystem. Plankton serves as a vital food source for many marine animals, impacting the food chain and ecosystem dynamics. Additionally, changes in plankton populations can influence nutrient cycling, water quality, and even climate regulation in the oceans. However, an overabundance of certain plankton species, like harmful algae, can lead to harmful algal blooms, which can have detrimental effects on marine life and ecosystem health. Therefore, while increased plankton can have positive aspects, it's crucial to monitor and maintain a balance to prevent any negative consequences.

OR

From handouts

Planktivore populations can impact the abundance and community composition of planktonic species through their predation pressure, and planktivore migrations facilitate nutrient transport between benthic and pelagic habitats.

• Planktivores are an important link in marine and freshwater systems that connect primary producers to the rest of the food chain. As climate change causes negative effects throughout the

global oceans, planktivores are often directly impacted through changes to food webs and prey availability.

• Additionally, harmful algal blooms (HABs) can negatively impact many planktivores and can transfer harmful toxins from the phytoplankton, to the planktivores, and along up the food chain.

26. what are dorsal fins and their functions in locomotion

Dorsal fins are located on the back or on the top of the fish, and aid the fish in sharp turns or stops, and assist the fish in rolling. Fish may have up to three distinct dorsal fins, known as proximal, middle , and distal dorsal fins, however several fish have just two dorsal fins with the middle and distal fins fused together.

27. Scutes?

In the sturgeon, ganoid scales are modified into body plates called scutes.

28. Difference between ganoid and ctenoid scales?

Ganoid scales are flat and do not overlap very much on the body of the fish. • They are found on gars and paddlefishes. • In the sturgeon, ganoid scales are modified into body plates called scutes.

Ctenoid & cycloid scales • Ctenoid scales are different than cycloid scales in that cycloid scales tend to be more oval in shape. • Ctenoid scales are more clam shaped and have spines over one edge. • Cycloid scales are found on fishes such as eels, goldfish, and trout. • Ctenoid scales are found on fishes like perches, wrasses, and parrotfish. Some flatfishes, like flounder, have both cycloid and ctenoid scales.

29. Properties of good fish pellet

For the farmers, the pellets' physical quality is of utmost significance.

- The pellet must have the proper sinking rate,
- Be able to resist storage,
- And not crumble or release fat.
- The meal must have the necessary nutritional composition,
- Be easily digested, and
- Attractive to the fish.

30. Feeding Habit of carps

Carps are omnivorous fish, meaning they eat both plant and animal matter. Their diet typically includes algae, aquatic plants, insects, crustaceans, and small fish. Carps are bottom-feeders, so they often scavenge for food at the bottom of ponds, lakes, or rivers. They have specialized pharyngeal teeth that help them crush and grind their food. Carps are known to be opportunistic feeders, consuming a wide variety of food sources depending on availability.

LONG QUESTIONS(5)

1. General characteristics of Fish:

- Fish are aquatic, gill-bearing animals
- That lack limbs with digits.
- Most fish are cold-blooded,
- Allowing their body temperatures to Vary as ambient temperatures change.
- •Their body is divisible into head, trunk and tail. Neck is absent.
- It has a spindle shaped body. It is helpful in swimming.
- On the head a pair of nostrils are present internal nostril are absent. • In Dipnoi Internal nostrils are present
- On the head a pair of eyes is present.
- On the lateral sides of the body LATERAL LINE SENSE organs are present. They detect the pressure changes of water.
- Kidneys are mesonephrlc. •
- Urinary bladder is absent •
- The scales of fish originate from the mesoderm (skin);

2. Muller's classification (1844)

• Earliest classification was proposed by J. Muller (1844) who proposed six subclasses in class Pisces.
• ***Class PISCES Subclasses:***

1) DIPNOI (Lung fishes);
2) TELEOSTEI (Bony fishes);
3) GANOIDEI (Polypterus, Amia, Lepidosteus);

4) ELASMOBRANCHI (Cartilagenous fishes);
5) MARSIPOBRANCHI (Cyclostomes);
6) LEPTOCARDII (Amphioxus).

3. Berg's classification (1940)

L.S. Berg classified fishes in to

• seven classes and separated lampreys and • hagfishes into a separate class.

• Berg included all the living and the fossils fishes • within a series-Pieces and• subdivided the series into seven major classes.

Series PlSCES

1) 3 Extinct Classes (Pterichthys, Coccostei, Acanthodi)
2) ELASMOBRANCHI (Cartilagenous fishes)
3) DIPNOI (Lung fishes)
4) HOLOCEPHALI (Chimaera)
5) TELEOSTOMI
    i. Crossoptergii (Fringe-Finned)
    ii. Actinoptergii (Ray-Finned)

4. Classification of Carp (Cyprinus carpio)

- Domain: Eukarya
- Kingdom: Animalia
- Phylumn: Chordata
- Class: Actinopterygii
- Order: Cypriniformes
- Family: Cyprinidae
- Genius:Cyprinus
- Species: *C. carpio*

5. Adaptation of gills for efficient gas exchange

1. Thin (Flattened cell) • Short diffusion distance

2. Permeable membranes • Allow gases to diffuse

3. Good vascular supply • Maintain concentration gradient

4. Large surface area • Allow more diffusion of gases

6. Main types of caudal fins in bony fish..

The caudal fin is known commonly as the tail fin. It is the primary appendage used for locomotion in many fishes. The main types of Caudal fins found in bony fish are:

- Homocercal shape tail
- The Heterocercal tail
- Round tail
- Truncated (triangular) tail
- Lunate (moon shaped) tail
- Forked tail

7. Types and function of pectoral fins.

The pectoral fins are vertical and are located on the sides of the fish, usually just past the operculum. • Pectoral fins are similar to human arms, which are found near the pectoral muscles. Many fish, such as reef fish like wrasses, use their pectoral fins for locomotion, holding position and swimming.

1) ***Fringe-like pectoral fins*** Probing substrate
2) ***Spiny pectoral fins*** Propping on substrate
3) ***Hand-like pectoral fins*** Crawling on substrate
4) ***Wing-like pectoral fins*** Soaring and swimming
5) ***No pectoral fins*** Snake-like swimming
6) ***Normal size pectoral fins*** Maneuvering

8. Lateral line system functions

Lateral line system: Detect motion or vibrations in water

The lateral line system is a sensory system found in primarily aquatic vertebrates like lampreys, fish, and amphibians. These animals have special mechanoreceptors called lateral-line organs in their outer skin (epidermis). These organs are sensitive to small water movements, especially those created by other animals moving in the water. The lateral line system helps these aquatic vertebrates detect

changes in water pressure and movement, allowing them to sense their surroundings and potential prey or predators.

9. Fish mouth morphology

❖ Mouth Type of fish

(i) ***Long and skinny bill*** •Tweezer like • Poking into crevices
(ii) ***Downward orientation*** • Oriented mouth • Useful to suck food from bottom • E.g., Janitor fish
(iii) ***Large mouth*** • Tearing large prey
(iv) ***Beaklike mouth*** • -used to graze on small algae growing on hard surfaces • E.g Parrot fish
(v) ***Upward orientation*** • Used to capture prey on water surface

❖ Mouth Position of fish

Most fish mouths fall into one of three general types:

(i) ***Superior***, or sometimes called ***supra-terminal, mouths*** are upturned. • The superior mouth is oriented upwards, and the lower jaw is longer than the upper jaw. • Archers, half-beaks, and hatchet fish are all examples of species of aquarium fish that have a superior mouth.
(ii) ***Terminal mouths*** point straight forward and are the most common mouth type. Terminal mouths are located in the middle of the head and point forward. Both jaws are the same length. More fish have this mouth type than any other.• Most barbs, cichlids, gouramis, and tetras have terminal mouths.
(iii) • ***Inferior***, ***or sub-terminal, mouths*** are turned downward. • Also called a sub-terminal or ventral mouth, the inferior mouth is turned downward. The lower jaw is shorter than the upper jaw, and the jaw will often be protrusible. • Most members of the catfish family have inferior jaws, and many of them also have a sucker mouth as well.
(iv) ***Protrusible mouth*** • A protrusible mouth allows a fish to extend its reach when attempting to snatch prey or food particles. • For example, kissing gourami uses its protrusible mouth to defend territory against others of the same specie.

10. Difference between telencephalon and diancephalon

Prosencephalon gives rise to two parts:

❖ Telencephalon (anterior part of the brain)

Mainly concerned with the receptor and conduction of smell. Telencephalon is divided into two regions:

1) Area ventralis:
2) and area dorsalis

Ventral area receives secondary olfactory fibers and it is a constant structure in all bony fishes • In adult fishes telecephalon may be formed by invagination and inversion and plays important role in the reproductive behavior of fishes.

❖ Diencephalon (the sight area of the brain)

• It is divides into three parts

1) Epithalamus
2) Hypothalamus
3) thalamus.

11. Characteristics of Sub class Holocepheli

• These fishes move by using sweeping movements of their large pectoral fins.

• They are deep sea fish with slender tails, living close to the seabed to feed on Benthic invertebrates.

• They lack a stomach

• Their food moving directly into the intestine.

• ***Example***• Rat fish

12. Chondrostean

☐ Chondrostean are thought to be the most primitive of the ray-finned fishes.

☐ They were at their greatest diversity during the Paleozoic and Mesozoic eras.

☐ Ancestral chondrostean had a bony skeleton, but living members, the sturgeons and paddlefishes, have cartilaginous skeletons.

☐ Chondrostean also have a tail with a large upper lobe.

13. Characteristics of Osteichthyes

• With more than 29,000 species of bony fishes, it is the largest class of vertebrates.

• It includes both marine and freshwater fishes, most of the commercially used fishes are included in this class.

• The size ranges from less than 8 mm in Paedocypris progenetica, which is also the smallest known vertebrate to 4 m and weigh about 1500 kg in the ocean sunfish (Mola mola).

• Their endoskeleton is made up of bones.

• Bony skeleton

• Operculum cover gills opening

• 20,000 species

• Present in Freshwater and saltwater

• Pneumatic sacs functions as lungs or swim bladder

• Bone in their skeleton, bony operculum covering the gills openings, lungs or swim bladder.

• Lateral line system

Fins: Dorsal, pectoral, caudal, pelvic, anal.

• Paired and median fins are present, which are supported by long rays of cartilage or bone. Fleshy lobed fins are present in sarcopterygians.

• These types of fins are supported by bones having joints.

• These are adapted to live at the bottom of the sea.

• First fossils of bony fishes are from late Silurian deposits.

14. Difference between Actinopterygii and Sarcopterygii.

1) Subclass Actinopterygii

• (Ray-finned fish) • Their fins lack muscular lobes that is why called as ray-finned fishes. • They usually possess swim bladders, gas filled sacs along dorsal wall of the cavity that regulate buoyancy. • Paired fins supported by dermal rays • Base of fins is not muscular • Tail fins almost equal to upper and lower lobes • Blind olfactory • ***Examples*** • Ray fined fishes

2) Subclass Sarcoptergyii

• (Lobe-finned fish) • Have muscular lobes associated with their fins and usually use lungs for gas exchange. • Pneumatic sacs function as lungs or swim bladder • Paired fins with muscular lobes • Some bony fishes are hermaphrodites and a number of species exhibit parthenogenesis • Fertilization is usually external • Development is usually oviparous (egg laying) but can be ovoviviparous or viviparous • ***Examples*** • Lungfishes, coelacanths

15. Super class Agnatha Characteristics

• No jaws

• No paired appendages

• Persistent Notochord

• Cartilaginous skeleton

• All living and most extinct

• Agnatha do not have an identifiable stomach or any appendages.

• Two semicircular canals

• Hag fishes, one semicircular canals

• Ostracoderm belong to several classes

• 15cm long

• Fossils: with water scorpions

• Sluggish

• Body armor for defense

• Bottom dwellers

• Filter feeders

• Crack Gastropods shells and Arthropod exoskeleton

• Annelids feeders

• The Agnatha are ectothermic or cold blooded,

• with a cartilaginous skeleton,

• and the heart contains 2 chambers.

• Fertilization and development are both external.

• There is no parental care in the Agnatha class.

16. Note on Class MYXINI

• Considered as most primitive fishes

• Known for its slimy secretions

• Cartilaginous skeleton

• Hagfish have elongated,

• eel-like bodies, and paddle-like tails.

• Sucker like mouth

• Mouth with four pairs of tentacles

• Five to fifteen pairs of pharyngeal slits

• Olfactory sacs apart to mouth

• Live in mud and sand of marine water.

• Hagfish are typically about 50 cm (19.7 in) in length.

• They are scavengers

• (enter in the large fish through mouth, eat internal parts leaving on skin and bones)

• Reproduction: rarely known

Example • Hag fishes

17. Respiration in hagfish

☐ A hagfish generally respires by taking in water through its pharynx and bringing the water through the internal gill pouches, which can vary in number from five to 16 pairs, depending on species. ☐ The gill pouches open individually, but in Myxini, the openings have coalesced, with canals running backwards from each opening under the skin, uniting to form a common aperture on the ventral side known as the ***branchial opening*** ☐ The esophagus is also connected to the left branchial opening, which is therefore larger than the right one, through a pharyngocutaneous duct (esophageocutaneous duct), which has no respiratory tissue. ☐ Hagfish also have a well-developed dermal capillary network that supplies the skin with oxygen when the animal is buried in anoxic mud, as well as a high tolerance for both hypoxia and anoxia, with a well developed anaerobic metabolism. ☐ The skin has also been suggested to be capable of cutaneous respiration.

18. Ciguatera poisoning

• Ciguatera poisoning is due to the consumption of fishes contaminated with ciguatoxin, a toxin which acts as a sodium channel agonist. • This toxin is produced by a microscopic marine dinoflagellate Gambierdiscus toxicus and through the food chain it reaches the fish, mainly the reef fishes. • The toxin accumulates mainly in the liver and testis of the fish as well as other viscera, while the flesh (muscle) of the fish is much less affected. • Cooking or storage of the fish has no effect on the toxicity. • Some of the fishes which have been implicated with ciguatera poisoning are groupers, barracuda, amberjack, sturgeon fish.

19. Osmoregulation in Fishes

❖ ***Freshwater Fishes:***

Freshwater fishes live in water with a much lower concentration of salts than they require inside their bodies. As a result, osmosis tends to promote the absorption of water into their bodies. To counteract this, their kidneys efficiently excrete water, and their gills reabsorb salt from the water. This helps maintain a stable internal environment and prevents excessive water uptake.

❖ ***Marine Fishes:***

Marine fishes, on the other hand, live in an environment where the water surrounding them has a greater concentration of salts than they can tolerate inside their bodies. To combat this, their kidneys and gills work together to remove excess salt. They produce a small amount of concentrated urine to eliminate salt, and their gills have special cells that actively eliminate salt. This helps maintain a stable internal environment and prevents dehydration.

20. How predatory fish adapted to capture its prey?

Fish that eat large animals as food are called predatory fish. • They have acquired some general adaptations to take food.

• They have well-developed grasping and holding teeth, such as many sharks (Elasmobranchii), Barracuda (Sphyraena), Pike (Esox), and Gers (Lepisosteus). • Their intestines tend to be smaller in size than herbivores.

• Many predatory fish such as bluegill (Pomatomus saltarix) and many deep sea fish actively hunt their prey.

• Other fish, such as the grouper (Epinephelus), often sit in wait till an animal passess and then durt out to grasp it.

• Angler fishes (Lophidae and Antennariidae) have developed an anterior ray of the first dorsal fin into a lure to attract their prey.

• The Southeast Asian archer fish (Toxotes jaculata) preys on insects by spitting on nearby trees. • This fish has a well-developed aerial viewing system.

• Some predatory fishes hunt by sight.

• Other fish, especially many sharks (Squaliformes), nocturnal fish such as bullheads (Ictalurus) and Morays (Muraenidae), catch the prey by ensuring the position of the prey with the help of smell, taste, touch and lateral line organs.

21. Differentiate between Herbivorous, Carnivorous, Omnivorous & Voracious Fishes/type of food of the omnivorous fishes/What kind of food do herbivorous fish eat?/What type of food is recommended for carnivorous fish.

Carnivores • Fish exclusively consume animal matters as food. While they will never harm plants. Their digestive track is developed to handle large pieces of meat and • Have room for a whole fish in their stomach. Carnivores are happiest when they are fed live ***food*** like worms. • Earthworms • Red worms, Shrimps • Oysters • Clams and other fish. . ***Examples*** of carnivorous fishes – salmon – billfish – barracuda – pike – walleye – perch – shark

Herbivores • Fish exclusively consume plant matters as food. Herbivorous fish most frequently graze, • – and whether or not they receive regular food, • – They nip at plant life. Their intestine break down the food because they do not have true stomach as in carnivorous fishes. • Herbivorous fish feed on the algae that grow on corals and compete with them for light and oxygen. • By eating the algae, these tiny fish play a big role in the survival of the entire ecosystem. ***Recommended food*** for the herbivores would be: • – Vegetable matter • – Peas • – Potatoes • – Cucumber • – Sea Food • – Fresh algae • ***Examples*** • Parrotfish • Damselfish • Surgeonfish

Omnivores • Fish consume both plant and animal matters as food. With the most versatile stomachs, omnivores' digestive tracts can derive nutrients from a variety of dietary options. Omnivore fish are destructive to plants and • – other small animals because they will feed just about anything. • – Additionally, they are voracious eaters. Omnivorous fish ***eat*** almost anything from vegetable and plant matter to insects, • – crustaceans and meat proteins • – a variety of seaweed, • – Shellfish • – plankton • – shrimp ***Examples*** Angelfish. ... • – Arowana. ... • – Cherry Barbs. ... • – Catfish. ... • – Discus. ... • – Goldfish. ... • – Guppies. ... • – Mollies.

Voracious Fishes • Fish consuming or craving large quantities of food. • Having or marked by an insatiable appetite for an activity or pursuit; greedy. • voracious predators sucking body fluids of many insects or blood of fish , tadpoles and frogs . • ***Recommended food*** for the voracious would be: • – Shark is an active, voracious predator of fish and invertebrates. • – Eels are various carnivores and consume a variety of fishes. • – and invertebrates such as worms, crayfish, snails and insects. • ***Examples*** • Piranhas • Barracuda • Pike

SHARED OBJECTIVE

1. Swedish naturalist _ is known as father of icthyology. *Peter artedi*

2. Gnathostome diversity comprises roughly ....*60,000 species*

3. Shark is about ..... In length. *15m*

4. A *protrusible mouth* allows a fish to extend its Reach when attempting to snatch prey or food particles.

5. lateral line system consists of _ or more sensory organs. *100*

6. In advanced teleosts, especially the acanthopterygians, the bladder has lost its connection with the digestive tract, a condition called.. *Physoclistic*

7. Fish belong to phylem. *Chordata*

8. Fish class maxini has..... Pair of tentacles *4*

9. *Piscivores* A piscivore is a carnivorous animal that eats primarily fish.

10. Herbivory is one of the most important processes in maintaining ecological balance on the *Mesoamerican Reef*

1. systematics: *evolutionary relationship between organisms.*

2. *Peter Artedi:* father of ichthyology.

3. L.S Berg classifies the fishes into *7* classes.

4. True about class Aganatha: *no identifiable stomach and paired appendages.*

5. Hagfishes *class myxinoidea*.

6. *Head shield provide some lift though* and make cephalospidomorphi better swimmers.

7. whale sharks measure up to *15m* in length.

8. holocephali 'complete head'.

9. *'Compressiform'* body shape is well designed for making quick turns and quick bursts of speed over short distance.

10. *Beaklike mouth* used to graze on small algae growing on hard surfaces.

11. *Dorsal fins* are located on the back or on the top of the fish, and aid the fish in sharp turns or stops.

12. The *pelvic fins* sit horizontally on the ventral side of the fish, past the pectoral fins.

13. Barbels with taste buds and without cartilage *(Cyprinidae, Carps).*

14. At the front of *olfactory lobes* a pair of structure that receive and process signals from the nostrils via two olfactory nerves.

15. Lateral lines in some species contains *electroreceptors,* which can detect electric signals in water generated by other organisms.

16. Mouth shape and tooth structure vary greatly in fishes, *depending on the kind of food normally eaten*.

17. *Carps* have a greater tolerance of low oxygen levels, pollutants and turbidity than most native fish, and are often associated with degraded habitats, including stagnant waters.

18. *Voracious

Fish* consuming or craving large quantities of food.

19. Detrivores fish *consume decaying plants and animals matter* as food.

20. *Bottom feeders (dwellers)* are fish that inhabit the lower level of the aquarium o feed off the bottom.

Mcqs :

1.Fish flourished in —— period

2.Gaustatory system is linked with——

(Digestion , reproduction, 2 or optiona thy)

3.Spiracles in fish are used for ?

4.Lateral line system is concentrated at which region?

5.Lateral line system is associated with?

6.Pelvic fin is definition thi

7.Operculum ki definition thi

✨*700* species Elasmobranchs

✨ *Agnatha* ... Do not have an identifiable stomach or any appendages.

✨ *Beaklike mouth* ..used to graze on small algae growing on hard surfaces

✨ First jawed fishes appeared in the geologic record about .... *100 million* years

✨ Hag fishes, belong to Class ... *MYXINI*

✨ The circulatory, or blood vascular, system consists of ... (*the heart, the arteries, the capillaries, and the veins*)

✨ Example of omnivores fish is *Goldfish*

✨ Subclass Dipnoi contains.. *Lungfish*

Mcqs

1. Herbivores fishes food is not....crab
2. omnivores have teeth that may be blunt and flat
3. Hemichromis specie in Africa
4. The Southeast Asian archer fish (Toxotes jaculata) preys on insects by spitting on nearby trees.
5. Systematics is the study of the diversity of fish and the evolutionary relationships between these organisms
6. The scales of fish originate from the mesoderm (skin)
7. Cosmoid (lungfishes and some fossil fishes)
8. Marigal is bottom feeder

1.Fish belongs to phylum

2. what is the impact of introducing omnivorous fish in ecosystem involving fish population

3.Select a strategy to introduce both carnivorous and herbivorous fish in an aquarium in ecosystem

4.Ray finned fishes belong to which sub class

5.Which scales are called as Placoid scales

6.What is the purpose of head in cephalospidomorphii

Mcqs tough thy

1.Good fish peller ki property

2.cartileginous fish ki kitni class hain

3.laverious fish feeding habit

4.upward mouth is used for

5.fish jo large animals per attach krti hai

Bht lengthy statement waly mcqs thy tough thy

Length of hagfish

Lung fish belong to

1_✨ Hog Fishes Belong to class (Myxini)

2_ ✨Sub Classes Elasmobranchii contain (700 species)

3_✨ Sub Class Diponi Contain (Lung Fishes)

4- ✨Lateral line System consists Of More or Sensory organs (💯)

5- ✨Omnivores Fish (Gold Fish🐟)

🦈 Shark is about.....15 m in length.

🐠..... Are lung fishes....Dipnoi.

🐠 Gnathostomes contain.....60000 species.

🐠 The primary fin of the fish is ..... Caudal fins.

🐠 Greater proportion ....40-60 % of mass fish muscle is made up of locomotry muscle.

1-Modification of the Oparin ---------- Haldane biochemical model is (Hydrothermal Model)

2. Chromistans Includes all of the following, except.....(Protozoa)

3-Reptiles ancestors of mammals are (basal synapsids)

4-Isthmus of panama separated (atlantic and pacific ocean)

5-Order and succession of rocks is called...(lithostratography)

- First jawed fishes appeared in the geologic record about ___ years after Jawless fish came about. .....100 million .
- ____are located on the back or on the top of the fish, and aid the fish in Sharp turns or stops, and assist the fish in rolling..... Dorsal fins
- Class elasmobranchii contains..... 700 species
- Gnathostome diversity comprises roughly ___, which accounts for 99% of all living vertebrates, including humans..... 60,000 species
- Swedish naturalist _ is known as father of icthyology.... Peter artedi
- Example of omnivores fish is....goldfish
- Romer's classified fishes into three classes
- Have muscular lobes associated with their fins and usually use lungs for gas exchange....lobe finned
- Carp belong to the class... Actinopterygii

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